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SCHOOL OF
COMPUTER SCIENCE,
INFORMATION &
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Software Engineering
CSCI 5801

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KenKen Puzzle Creator

Introduction

This document describes the behavior of the KenKen puzzle creator, which is the front end of the software system being designed in this course.

KenKen is a numeric puzzle game similar in some respects to Sudoku. It uses a variable-sized grid, typically 9×9. Like Sudoku, each digit can only appear once in any row or column. However, the grid is divided into contiguous regions with various shapes. Each region contains a target number and possibly one of the four arithmetic operators. Initially, all of the individual cells are empty.

To solve the puzzle, digits must be placed in all of the cells such that the following conditions apply:

- Each digit only appears once across a row.
- Each digit only appears once in a column.
- If a region has no arithmetic operator:
  - It must consist of a single cell.
  - The target number is the digit that fills in the cell.
- If the region's arithmetic operator is subtraction:
  - The region must consist of two cells.
  - The difference of the two digits must be the region's target number.
  - The digits may appear in either order.
- If the region's arithmetic operator is division:
  - The region must consist of two cells.
  - The quotient of the two digits must be the region's target number.
  - The remainder after division must be zero.
  - The digits may appear in either order.
- If the region's arithmetic operator is addition or multiplication:
  - The sum or product of the digits must be the region's target number.
  - Digits may appear more than once in the region.
  - The digits may appear in any order that does not violate other constraints.

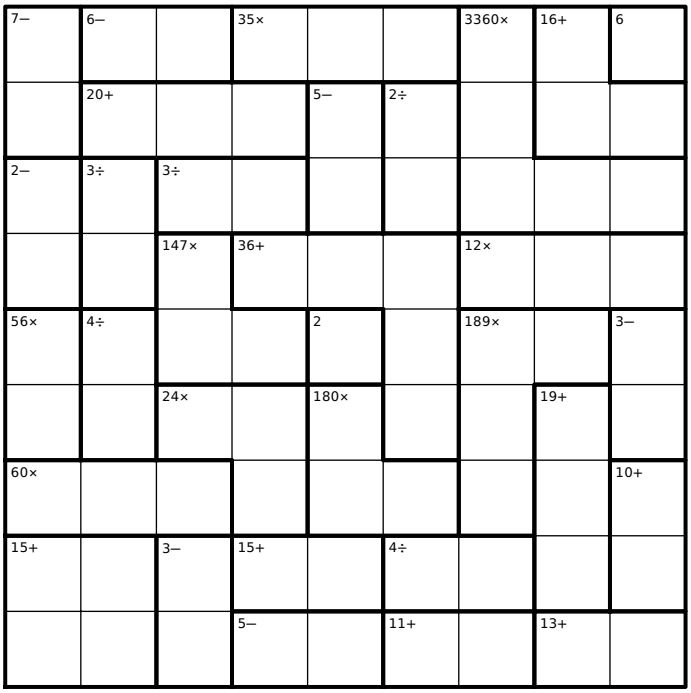


Figure 1: A sample KenKen puzzle

Goal

The goal of this part of the project is to construct a program that allows a user to create an empty KenKen puzzle, then send a description of the puzzle to another subsystem that will begin the solution process.

Figure 1 shows a sample KenKen board:

The subsystem described in this document will allow a user to enter the information about each of the regions (also called *cages*) in the puzzle.

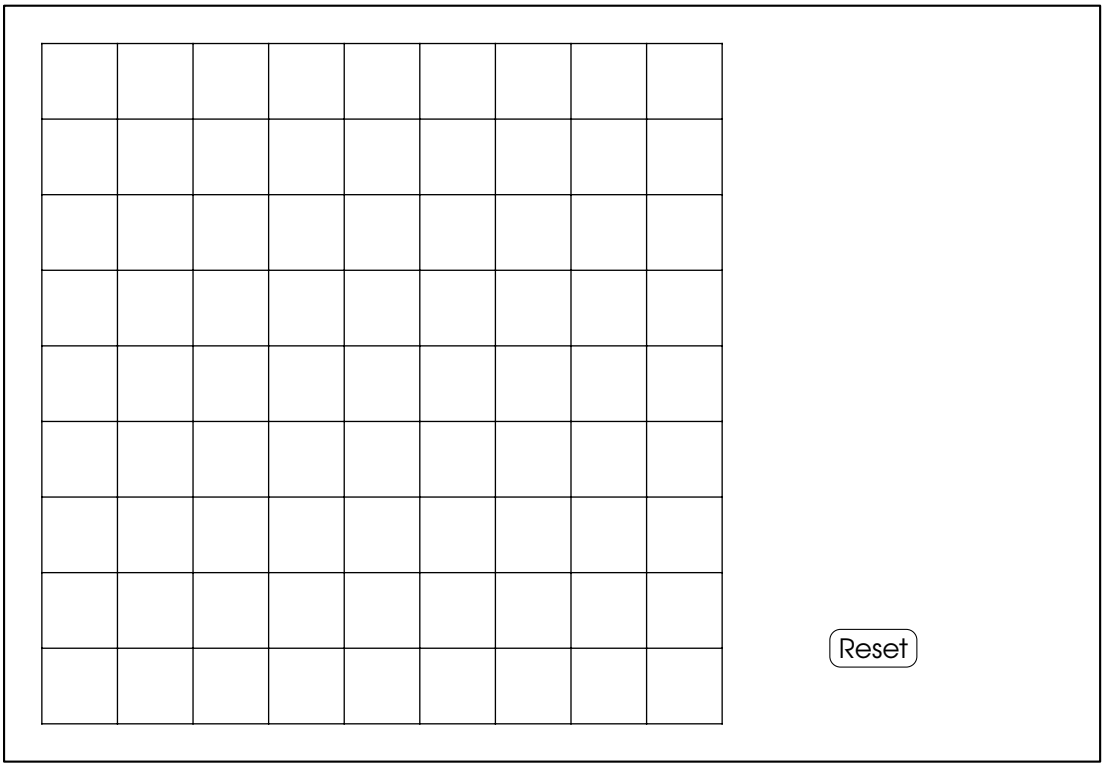


Figure 2: A starting KenKen puzzle with the reset button

Details

The starting state

Initially, the puzzle has no regions; it is an empty 9 × 9 grid. The following is a mock-up of what the software should display; your solution does not need to look completely identical to this, but it does need to display the proper elements and implement the proper actions.
Your solution should begin by displaying an empty 9 × 9 grid to the user and allow the user to click on various elements; Figure 2 shows a possible view. The actions that should take place when elements are clicked are described below.

Clicking on an unselected cell that is not part of a region

The intent of clicking on such a cell is that the user is building a region. The action to take depends upon the current state of the puzzle.

If no other cell is selected

In this case, highlight the cell. The software should also display and/or enable the following elements:

- A "Create Region" button
- A "Cancel" button
- A field to enter the region's target number

- An element to select the region's arithmetic operator. The exact form of this is at your discretion, but should be something that does not require a keyboard.

If another cell has already been selected

If the current cell is adjacent to a highlighted cell, then highlight the cell. Otherwise, ignore the click; highlighting the cell would create a region that is not contiguous.

If a region has already been selected

Ignore the click. Once a region has been created, it cannot be edited, only deleted.

Clicking on the "Create Region" button

This button should only be enabled if each of the following is true:

- At least one cell not belonging to a region is highlighted
- A number has been entered in the target number field
- An arithmetic operator (possibly "none") has been selected

If the button is clicked while enabled, create a region consisting of all highlighted cells, the given target number and arithmetic operator. Then, redraw the grid with the new region, and disable and/or hide the "Create Region" and "Cancel" buttons, the target number field and arithmetic operator widget. Clear any values set in the latter two items.

If all cells now belong to a region, display and/or enable a "Solve Puzzle" button.
Notes:

- This subsystem does not need to check that the target number is actually realizable. For example, a target of 10 is not possible with division or subtraction as it requires numbers larger than 9. That is handled by the second subsystem in the project.
- The new region should not be highlighted when it it redrawn.

If the "Cancel" button is clicked

The intent is to unselect all cells and cancel the creation of a region. Remove all highlighting and disable and/or hide the "Create Region" and "Cancel" buttons and the target number and arithmetic operation fields. The latter two items should also be cleared of any value that might have been set previously.

The target number field

You should ensure that any value entered here is a valid positive integer. Disallow negative numbers, real numbers or anything nonnumeric.

The arithmetic operator field

This allows the user to select a single arithmetic operator or "none." This can be a dropdown, a set of radio buttons or any other widget that allows the user to easily select one of the five possible values.

Clicking on a selected cell that is not part of a region

The user is creating a region but did not want to select this cell. Unhighlighting the cell risks splitting the tentative region into separate parts. There are two options for handling this click:

- Ignore the click if unselecting the cell would split the remaining highlighted cells
- Ignore the click regardless of whether or not it would cause a split

Handle this at your discretion.

Clicking on an unhighlighted cell that is part of a region

There are two possible courses of action to take, depending on the current state of the puzzle.

If no other cells are highlighted

This subsystem is easiest to construct if regions cannot be edited once created. With this in mind, the only action that could be carried out on a selected region is to delete it.

If such a cell is clicked, highlight all cells in the region, then display and/or enable a "Delete Region" button.

If any other cells are highlighted

If any other cells are highlighted, whether or not they are currently in a region, ignore the click.

Clicking on the "Delete Region" button

This can only be done if a single region is currently highlighted. Delete the region, redraw the grid without the region and disable and/or hide the "Solve Puzzle" button if it is currently enabled.

Clicking on a highlighted cell that is part of a region

The intent is to toggle the selection status of the region. Unhighlight all cells in the region and disable and/or hide the "Delete Region" button.

Clicking on the "Solve Puzzle" button

If this is clicked while enabled, then the entire puzzle should be sent to the second subsystem to generate candidate solutions. In addition, the empty puzzle should be saved to a file. For both destinations, use the data format described in the Data Interchange Format document.

The "Reset" button

One final requirement is to have a reset button. This will reset the puzzle to the original empty state with nothing highlighted and no other buttons active.

Platform

As long as the subsystem meets the above requirements, the platform does not matter. You can design the software as a web app or desktop app. Use any appropriate programming language.

Examples

The examples on the following pages illustrate possible views of the user interface at various stages of operation. These are for illustration purposes; your system may look somewhat different from these.

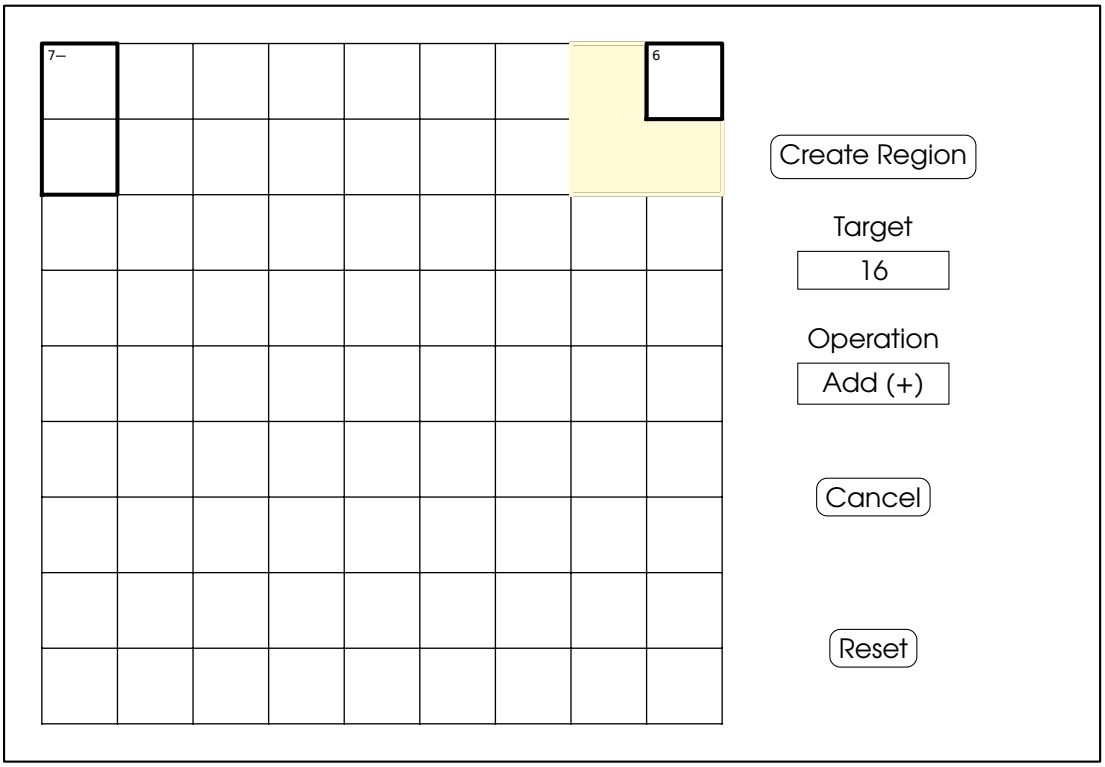


Figure 3: Highlighting cells for a new region

Example 1 — Highlighting cells

Figure 3 shows an example of highlighting three cells in the top-right corner in order to create a new region.

Note that the create and cancel buttons are shown and active, and the target number field has a value as does the arithmetic operation.

Clicking on the top-right corner cell will do nothing, since it is already part of another region.

Clicking any of the other five cells adjacent to a highlighted cell will highlight that cell also.

Clicking any other cell will do nothing since it would not be adjacent to cells currently highlighted.

Clicking "Create Region" will create a new region with these three cells.

Clicking "Cancel" with turn off the highlighting and disable or hide the create, cancel, target and operation widgets.

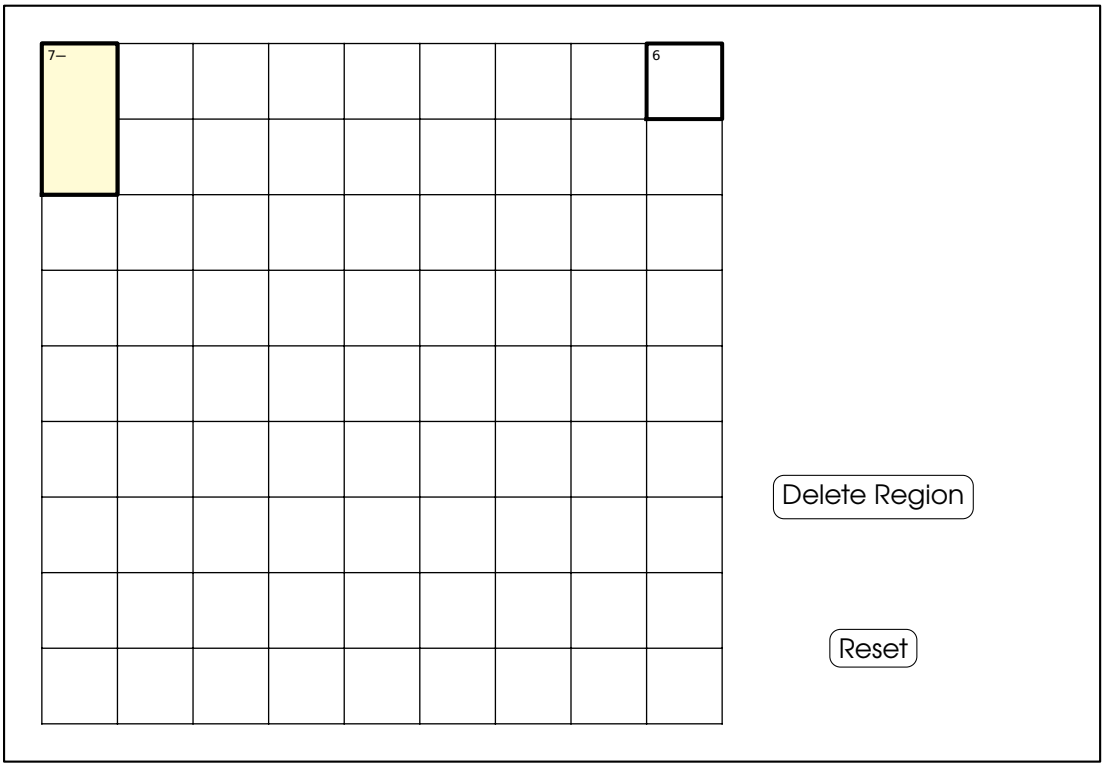


Figure 4: Highlighting a region

Example 2 — Highlighting regions

Figure 4 shows an example of highlighting the region in the top-left corner.

The user can click on the region again to unselect the region, or it can be removed by clicking the "Delete Region" button.

Clicking on any other cell or region will have no effect.

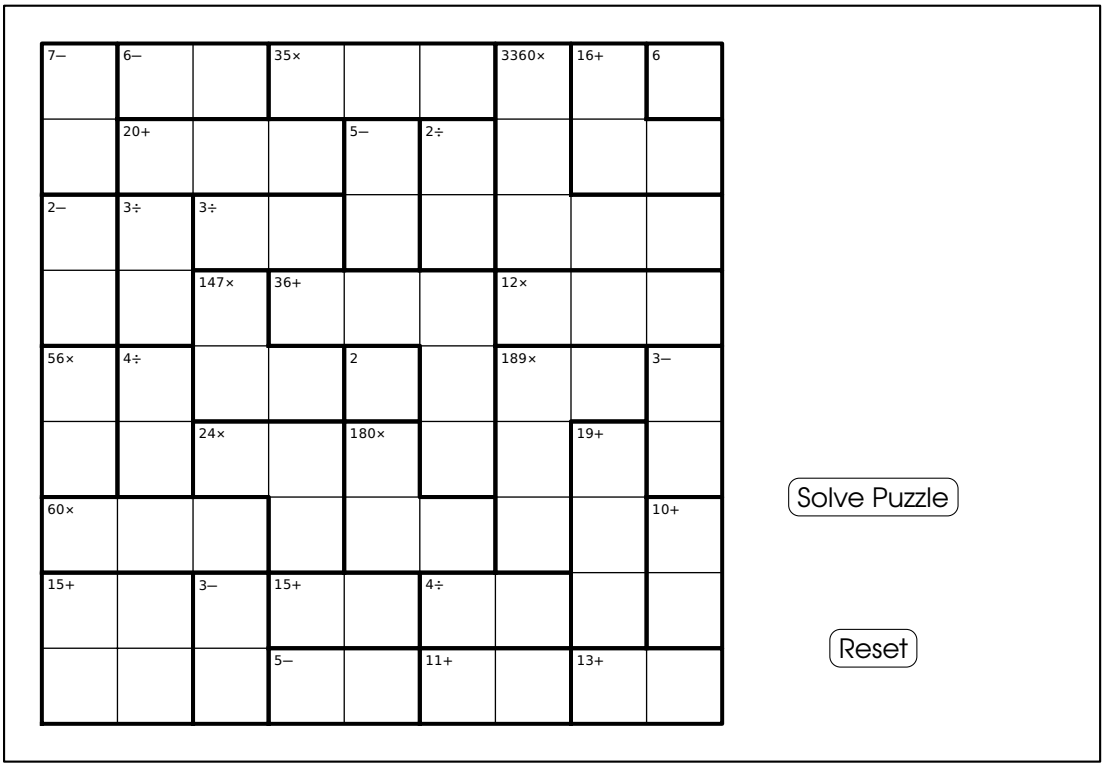


Figure 5: A completed puzzle

Example 3 — "Solving" the puzzle

Figure 5 shows an example of a complete puzzle ready for solving. Clicking on "Solve Puzzle" will send the puzzle to the solver subsystem and create an XML file.